# Policymakers' Uncertainty: The Risk-Management Channel

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# Monetary Policy and Uncertainty

Policymakers have long viewed uncertainty as central to decision-making

*"Uncertainty is not just an important feature of the monetary policy landscape; it is the defining characteristic of that landscape. . . As a consequence, the conduct of monetary policy in the United States has come to involve, at its core, crucial elements of risk management."* (Greenspan 2004)

*"The Governing Council bases its monetary policy decisions, including the evaluation of the proportionality of its decisions and potential side effects, on an integrated assessment of all relevant factors. In particular, it takes into account not only the most likely path for inflation and the economy but also surrounding risks and uncertainty"* (ECB Strategy Statement 2025)

Uncertainty also discussed in 2025 Fed's strategy review (Bauer et al 2025, Garga et al 2025)

# This Paper

**Research Question**

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Challenge: How to measure risk/uncertainty?

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Why should we care?

Can CBs replicate the approach to help their policy process?

How should we model CB decision making (e.g. to derive shocks)?

What should CBs communicate?

# Contributions

1. Framework for organizing potential effects of risk management
2. Use FOMC transcripts to measure balance-of-risk (BoR) perceptions at individual level.
3. Show that BoR perceptions have significant effect on:
   - 3.1 Policy decisions
   - 3.2 Romer-Romer (RR) and Aruoba-Drechsel (AD) shocks
   - 3.3 Individual policy preferences (in panel regression framework)
4. These effects arise from meetings where "insurance" is an expressed motive for policy.
5. Show that BoR perceptions also influence signals of future policy from the meeting.

# Outline

1. Framework

2. Data and Measurement

3. Balance of Risks and Policy Outcomes

4. Insurance Motives

5. Balance of Risks and Member-Level Preferences

6. Signalling about Future Policy

7. Macroeconomic Effects

8. Conclusion

# Standard Policy Rule

Policymaker solves $\min_{r_t} E[L(\pi_t, y_t) \mid r_t]$ given standard preferences

$$L(\pi_t, y_t) = (\pi_t - \pi^*)^2 + \lambda(y_t - y^*)^2$$

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Expected losses ($\mu_{\pi,t}(r_t)$ expected inflation; $\mu_{y,t}(r_t)$ expected output)

$$E[L(\pi_t, y_t) \mid r_t] = (\mu_{\pi,t}(r_t) - \pi^*)^2 + V_{\pi,t}(r_t) + \lambda(\mu_{y,t}(r_t) - y^*)^2 + \lambda V_{y,t}(r_t)$$

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Standard Taylor rule emerges from:

1. **Linear means**. $\mu_{\pi,t}(r_t) = a_{\pi,t} - b_\pi r_t \qquad \mu_{y,t}(r_t) = a_{y,t} - b_y r_t$
2. **No higher-moment policy effects**. $V'_{\pi,t}(r_t) = V'_{y,t}(r_t) = 0$

$$\hat{r}_t = \frac{b_\pi}{c}(a_{\pi,t} - \pi^*) + \frac{\lambda b_y}{c}(a_{y,t} - y^*), \qquad c \equiv b_\pi^2 + \lambda b_y^2.$$

This rule satisfies certainty equivalence.

# Risk Management as Alternative Scenarios

*"What does risk management actually mean? It entails thinking about what could go wrong with the forecast and then judging if policy should be adjusted from the baseline one way or the other in light of the alternative scenarios."* (Evans 2019)

Let $\omega_t \in \{0, 1\}$ where $\omega_t = 1$ is an adverse outcome with probability $p_t < 0.5$ and size $\Delta_t$, e.g.,

- $\pi_t = \tilde{\pi}_t + \omega_t \Delta_t$, with baseline $E[\tilde{\pi}_t \mid r_t] = a_{\pi,t} - b_\pi r_t \rightarrow$ Upside inflation risk
- $y_t = \tilde{y}_t - \omega_t \Delta_t$, with baseline $E[\tilde{y}_t \mid r_t] = a_{y,t} - b_y r_t \rightarrow$ Downside growth risk

Expected loss over probability-weighted scenarios is

$$E\left[L(\pi_t, y_t) \mid r_t\right] = (1 - p_t(r_t))\, \ell_{0,t}(r_t) + p_t(r_t)\, \ell_{1,t}(r_t)$$

where $\ell_{\omega,t}(r_t)$ is the expected loss conditional on state $\omega_t = \omega$.

# Exogenous Tail Probability (Certainty Equivalence)

For concreteness, consider an upside inflation risk and linear-quadratic environment.

Suppose that $p_t'(r_t) = 0$ so that tail probability is exogenous to policy. Optimal policy is

$$\hat{r}_t = \frac{b_\pi}{c}\left(a_{\pi,t} + p_t\Delta_t - \pi^*\right) + \frac{\lambda b_y}{c}\left(a_{y,t} - y^*\right),$$

which continues to satisfy certainty equivalence.

Risk management shifts policy due to mean-mode wedge (De Polis et al. 2026).

# Endogenous Tail Probability (Insurance)

Now suppose that $p_t'(r_t) \neq 0$ so that tail probability depends on policy. We call this **insurance**.

Can arise from several forces: financial instability (Woodford 2012, Svensson 2017, Boissay et al. 2025); disanchoring of inflation expectations (Goodfriend 1993); zero lower bound (Evans et al. 2015).

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Optimal policy is now

$$\underbrace{(1 - p_t(\hat{r}_t))\,\ell_{0,t}'(\hat{r}_t) + p_t(\hat{r}_t)\,\ell_{1,t}'(\hat{r}_t)}_{\text{fixed-probability tradeoff}} + \underbrace{p_t'(\hat{r}_t)\,[\ell_{1,t}(\hat{r}_t) - \ell_{0,t}(\hat{r}_t)]}_{\text{insurance motive}} = 0$$

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In the linear-quadratic case with upside inflation risk, this simplifies to

$$\hat{r}_t = \underbrace{\frac{b_\pi}{c}\left(a_{\pi,t} - \pi^* + p_t(\hat{r}_t)\Delta_t\right) + \frac{\lambda b_y}{c}\left(a_{y,t} - y^*\right)}_{\text{certainty equivalence rule}} + \underbrace{\frac{|p_t'(\hat{r}_t)|\Delta_t}{c}\left[\left(a_{\pi,t} - b_\pi \hat{r}_t - \pi^*\right) + \frac{\Delta_t}{2}\right]}_{\text{insurance wedge}}$$

# Empirical Insights

Policymakers react to economic conditions more aggressively than expected outcomes alone dictate.

Shows that costly events can have large policy effects even when $p_t$ is small (as long as $p_t'$ is not).

Later we distinguish our mechanism from others (e.g. robust control) by specifying detailed conditions under which insurance operates.

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# FOMC Transcripts

FOMC transcripts are publicly available from 1976 onwards, published with a five-year lag.

Verbatim, attributed, (nearly) complete record of policy deliberations. (Some missing material due to technical error or redaction due to sensitivity of the material.)

We focus on scheduled meetings, not conference calls.

Main sample covers October 1982—December 2019, 295 meetings in total.

105 unique FOMC members.

# Meeting Structure

FOMC meetings follow a regular structure in much of our sample. Core parts:

1. **Economy Round**
   - Staff presentation on economic and financial conditions
   - Q&A on staff presentation
   - FOMC member presentations on their assessment and outlook

2. **Policy Round**
   - Staff presentation on policy alternatives
   - Q&A on policy alternatives
   - FOMC members sequentially state and justify preferred alternative

# Measuring Risk Perceptions at the Individual Level via an LLM

Feed each FOMC member's economy round text into an LLM (Claude Sonnet 4) and ask for a risk assessment in each of two dimensions: inflation and growth.

Four categories: Upside, Balanced, Downside, Not Discussed

Coded view $BoR_{it}^{v} \in \{U, B, D, N\}$ for member $i$, variable $v \in \{I, G\}$, meeting $t$.

Query focuses on risk perceptions *around* individual forecasts, not disagreement with staff.

Most member-meeting statements are labeled as 'not discussed':

- 61.5% for inflation
- 52.3% for growth

# Example Quotations (Inflation)

*"We can't dismiss the possibility that compensation growth will drift upward, raising core inflation"*
Yellen, May 1996

*"I would also judge the risk to the inflation outlook as being biased to the downside"*
Lockhart, Sep 2010

*"The risks to the inflation outlook remain balanced"*
Bostic, May 2019

# Example Quotations (Growth)

*"I'm beginning to think our greatest problem may be an excessive increase in activity"*
Partee, Feb 1985

*"The risk to satisfactory economic performance... will remain skewed very much to the downside"*
Kohn, Mar 2008

*"Considerable risks still in place to growth, including gasoline prices, Europe and the deleveraging of European banks, continued stresses and tight credit in the U.S. banking system..."*
Bernanke, Mar 2012

# Meeting-Level Balance-of-Risks Measure

Define counts of FOMC members classified as upside/downside:

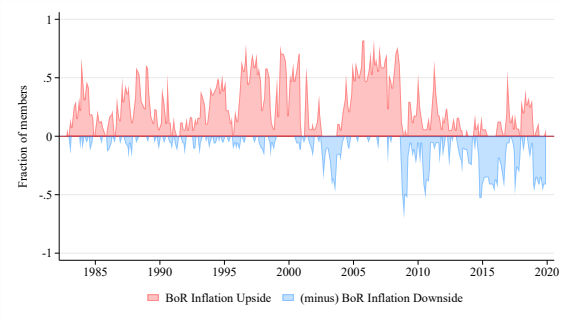
$$R^v_{U,t} = \sum_i \mathbf{1}(BoR^v_{it} = U), \qquad R^v_{D,t} = \sum_i \mathbf{1}(BoR^v_{it} = D).$$

Aggregate to meeting level upside & downside measures for variable $v$:

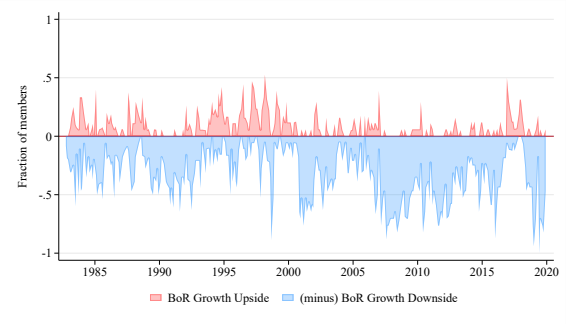
$$BoR^{(U),v}_t = \frac{R^v_{U,t}}{N_t}, \qquad BoR^{(D),v}_t = \frac{R^v_{D,t}}{N_t},$$

where $N_t$ is the number of FOMC members at meeting $t$.

# BoR Perceptions at the Meeting Level




Panel A. BoR perceptions for inflation




Panel B. BoR perceptions for growth

# Validation I: Summary of Economic Projections

From Oct 2007, and four times a year from 2008 onwards, FOMC members respond to a survey about economic conditions collated in the Summary of Economic Projections.

Includes question "Please indicate your judgment of the risk weighting around your projections" with three responses:

- Weighted to downside
- Broadly balanced
- Weighted to upside

Additional questions on member-specific point estimates of macro variables and optimal policy used as controls in regression analysis.

# Comparison to SEP

Member-meeting level: count (row %); 837 observations, 50 SEP meetings, 2007:10–2019:12

| | LLM | | | | | |
|---|---|---|---|---|---|---|
| | GDP growth risk | | | PCE inflation risk | | |
| SEP | Upside | Balanced | Downside | Upside | Balanced | Downside |
| Upside | 15 (60%) | 2 (8%) | 8 (32%) | 66 (89%) | 1 (1%) | 7 (9%) |
| Balanced | 34 (14%) | 68 (29%) | 136 (57%) | 78 (34%) | 79 (34%) | 72 (31%) |
| Downside | 2 (1%) | 10 (4%) | 240 (95%) | 5 (5%) | 2 (2%) | 101 (94%) |
| | SEP | | | | | |
| LLM Not discussed | 16 (5%) | 231 (72%) | 75 (23%) | 43 (10%) | 314 (74%) | 69 (16%) |

# Examples of Mismatch

**Kocherlakota Apr 2012, Inflation, LLM: Upside, SEP: Balanced**
"In this scenario, we might well see rapid increases in inflation and inflation expectations when labor markets seem weak by historical standards. . . This does not strike me as a risk that we should ignore."

**Yellen Dec 2013, Growth, LLM: Balanced, SEP: Downside**
"I now see appreciably diminished downside risks to the outlook for the labor market and more balanced risks to growth than in September."

**Bullard Sep 2016, Inflation, LLM: Downside, SEP: Upside**
"I do remain concerned that market-based measures of inflation expectations remain too low to be consistent with the 2 percent inflation forecast. This is a risk to the outlook that I have described."

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# Policy Outcomes with Extended Sample

We estimate baseline policy rule regressions, which we progressively augment with the balance-of-risk variables and controls:

$$\Delta_{(1d)} FFR_t = \alpha + \gamma' \mathbf{BoR}_t + \beta_1 FFR_{t-(1d)} + \beta_2 FFR_{t-1} + \delta' \mathbf{X}_t + \varepsilon_t. \quad (1)$$

Estimated at the meeting frequency, where $t$ indexes meetings.

Dependent variable is the one-day change in the policy rate over the 1982:10–2019:12 sample with Krippner shadow short rate used during the ELB period (2009:01–2015:10)

$\mathbf{BoR}_t = (BoR_t^{(U),I}, BoR_t^{(D),G}, BoR_t^{(D),I}, BoR_t^{(U),G})'$.

$\mathbf{X}_t$ controls comprise the Greenbook one-quarter-ahead ($q1$) CPI inflation forecast, the current-quarter ($q0$) real GDP growth nowcast, updates of one-quarter-ahead forecasts relative to the previous Greenbook, and trend inflation.

# Balance of Risk and Current Policy Action

| | FFR changes (pp), $\Delta_{(1d)}FFR_t = FFR_t - FFR_{t-(1d)}$ | | | | | | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| BoR Inflation Up | | 0.221*** | 0.196*** | 0.170*** | 0.184*** | 0.160*** | 0.166*** |
| | | (2.97) | (3.53) | (3.18) | (3.75) | (3.13) | (3.26) |
| BoR Growth Down | | -0.310*** | -0.191*** | -0.173*** | -0.156*** | -0.180*** | -0.189*** |
| | | (-3.63) | (-3.56) | (-3.59) | (-3.46) | (-3.59) | (-4.18) |
| BoR Inflation Down | | 0.114 | 0.0888 | | | | |
| | | (0.82) | (1.17) | | | | |
| BoR Growth Up | | 0.0877 | -0.0428 | | | | |
| | | (0.61) | (-0.34) | | | | |
| GB Controls | Yes | No | Yes | Yes | Yes | Yes | Yes |
| FFR lags | Yes | No | Yes | Yes | Yes | Yes | Yes |
| Trend $\pi$, Cons | Yes | No | Yes | Yes | Yes | Yes | Yes |
| Public uncertainty | No | No | No | No | No | Yes | No |
| Other horizons | No | No | No | No | Yes | No | No |
| Financial conditions | No | No | No | No | No | No | Yes |
| $\bar{R}^2$ | 0.36 | 0.20 | 0.41 | 0.42 | 0.44 | 0.44 | 0.43 |
| $\Delta R^2$ | . | | 0.061 | 0.058 | 0.056 | 0.060 | 0.061 |
| N | 295 | 295 | 295 | 295 | 295 | 245 | 295 |

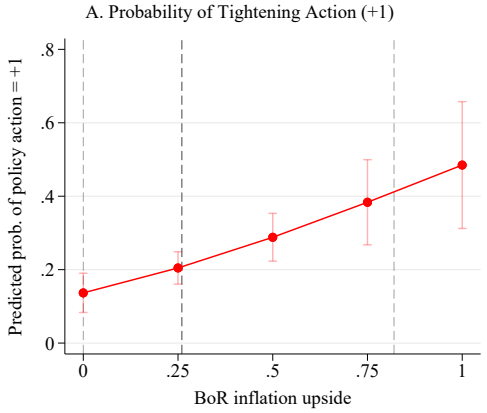
1-SD change in upside inflation BoR = 79% of a 1-SD change in the Greenbook 1Q-ahead inflation forecast coefficient.

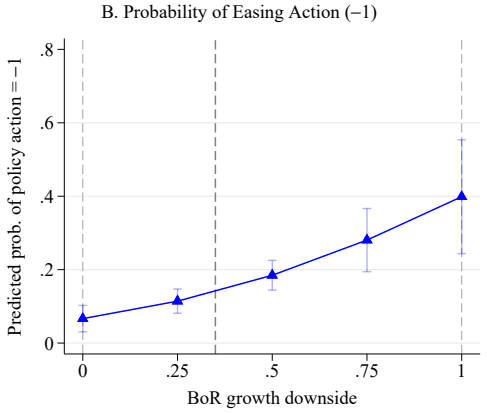
1-SD change in downside growth BoR = 59% of a 1-SD change in the Greenbook real GDP nowcast coefficient.

# BoR and Policy Action Probabilities

Create a discrete policy action $D_t^{Act} = \{-1, 0, +1\}$

Estimate an ordered probit on BoR variables, with the baseline controls.



# Monetary Policy Shocks

1. Romer and Romer shocks orthogonalize Fed Funds Rate with respect to Greenbook controls.

2. Aruoba and Drechsel shocks: further orthogonalize with respect to textual sentiment indicators extracted from staff materials (color books).

   $\Longrightarrow$ explicitly motivated by potential divergence between modal numeric Greenbook forecasts and conditional mean forecast, driven by asymmetric balance of risks.

# Balance of Risk and Identified Monetary Policy Shocks

| | RR shock (pp) | | AD shock (pp) | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| BoR Inflation Up | 0.19*** | 0.19*** | 0.055*** | 0.057*** |
| | (3.17) | (3.47) | (3.10) | (3.58) |
| BoR Growth Down | -0.19** | -0.15** | -0.023 | -0.027 |
| | (-2.35) | (-2.00) | (-1.15) | (-1.47) |
| BoR Inflation Down | -0.17 | | -0.0060 | |
| | (-1.07) | | (-0.13) | |
| BoR Growth Up | -0.20 | | 0.013 | |
| | (-1.56) | | (0.31) | |
| Sample | 1982:11–2007:12 | | 1982:11–2008:10 | |
| $\bar{R}^2$ | 0.099 | 0.091 | 0.058 | 0.067 |
| N | 201 | 201 | 207 | 207 |

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# Measuring Insurance Reasoning

To separate meetings by the presence of insurance motives, we design a new LLM query that operates on policy round text at the meeting-speaker level.

Require two conditions:

1. Member articulates a specific, low-probability adverse scenario
2. Scenario directly motivates a policy response (via current or future action)

Two-stage classification:

- Stage 1: broad patterns of insurance reasoning
- Stage 2: recovers two specific scenarios missed by stage 1 (entrenchment of inflation expectations; acting too slowly at the onset of a downturn)

Overall: 25% (1,337 of 5,252) member-meeting statements contain insurance reasoning – lower bound

# Meeting-Level Classification

From speaker-level to meeting-level insurance variables:

1. Insurance share = fraction of non-dissenting members invoking insurance as justification for the current action
2. Insurance meeting dummy = 1 if:
   - Chair invokes insurance reasoning, *or*
   - Share of members invoking it exceeds 25% (75th percentile of cross-meeting distribution)

Of 298 meetings, 104 classified as insurance-driven; mild persistence ($P(1 \to 1) = 0.47$)

# Insurance at the Fed: September and November 1998 = Downside growth

*Insurance(Share)= 0.3 / 0.27, Insurance(Chair)= 1 in both meetings*

Both meetings are classified as insurance meetings, and our downside-growth risk measure is high in each ($BoR_t^{(D),G} = 0.89$ in September and $BoR_t^{(D),G} = 0.57$ in November).

Greenspan: "it is conceivable that we may end up viewing this action not as the first in a series of moves but as an insurance premium."

Governor Meyer supported the easing and explicitly cited that it could be justified as "providing insurance against a bad draw from the unpleasantly fat tail of the probability distribution of outcomes".

In November, Greenspan judged that the economy was doing reasonably well, with only modest deterioration but: "there could be a far more deleterious outcome ahead" concluding that "a little insurance at this stage is probably wise."

Governor Ferguson: "we can afford to buy some insurance at this time. I think the premiums are manageable. I think the risks of not buying some insurance are just too great for my taste."

# Insurance at the Fed: December 1988 = Upside inflation

*Insurance(Share)= 0.2, Insurance(Chair)= 1*

Upside inflation concerns had built ($BoR_t^{(U),I} = 0.61$), despite policy having firmed over the year.

Greenspan's argument turned on credibility and the danger of inflation becoming entrenched.

He contended that the market's perception of Fed resolve was itself suppressing wage acceleration that "would occur otherwise" and that continued firmness was needed to preserve that credibility.

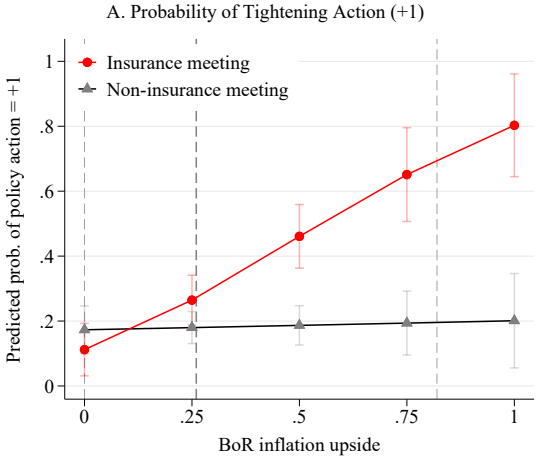
Greenspan stressed that he felt the costs of tightening in terms of hurting the economy were small:

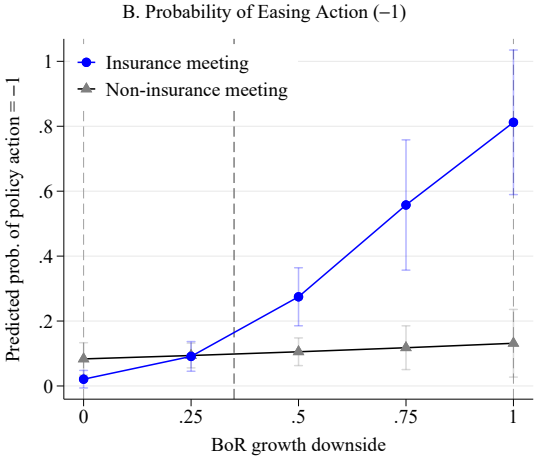
> *"...I think that with the contemplated tightening that we've been discussing the risk of tilting this economy over at this stage is exceptionally small. Moving forward and finding that we made a mistake is a possibility, but I think the cost of such a mistake is really much smaller than the cost of doing nothing in the face of what basically is going on."*

# Balance of Risk and Shocks: Insurance vs. Non-Insurance Meetings

| | Insurance = 1 | | | Insurance = 0 | | |
|---|---|---|---|---|---|---|
| | (1) $\Delta_{1d}FFR$ | (2) RR | (3) AD | (4) $\Delta_{1d}FFR$ | (5) RR | (6) AD |
| BoR Inflation Up | 0.28** | 0.30** | 0.13*** | 0.082 | 0.11 | 0.012 |
| | (2.49) | (2.60) | (4.08) | (1.34) | (1.63) | (0.58) |
| BoR Growth Down | -0.31*** | -0.47*** | -0.076** | -0.085 | -0.015 | 0.027 |
| | (-3.30) | (-3.26) | (-2.19) | (-1.54) | (-0.15) | (1.22) |
| BoR Inflation Down | -0.0093 | -0.10 | 0.071 | 0.070 | -0.072 | -0.0094 |
| | (-0.06) | (-0.38) | (0.90) | (1.03) | (-0.38) | (-0.21) |
| BoR Growth Up | 0.16 | -0.060 | 0.15* | -0.091 | -0.26** | -0.043 |
| | (0.56) | (-0.20) | (1.98) | (-0.98) | (-2.16) | (-1.29) |
| Controls | Yes | No | No | Yes | No | No |
| $\bar{R}^2$ | 0.62 | 0.34 | 0.34 | 0.27 | 0.020 | 0.013 |
| N | 100 | 73 | 78 | 194 | 127 | 128 |

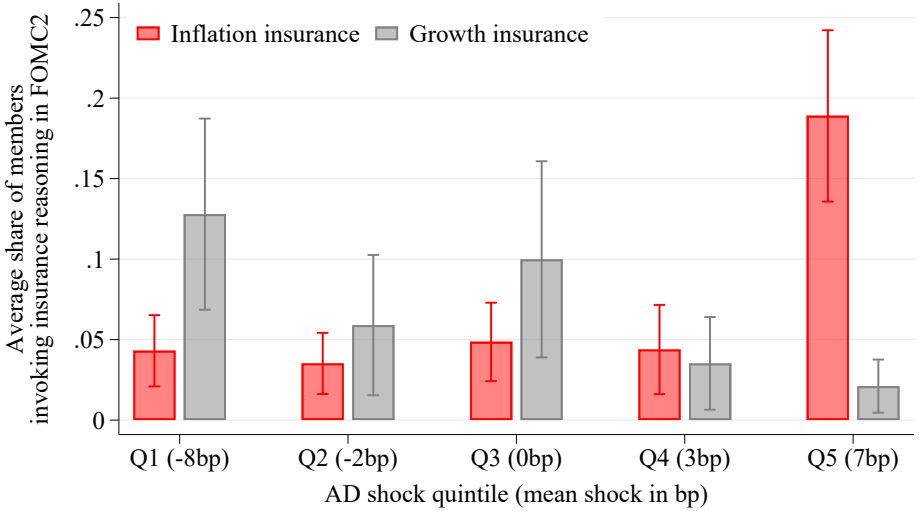
# Probability of Policy Action: Insurance vs. Non-Insurance Meetings



# Insurance Reasoning and the Size of Monetary Policy Shocks

Romer & Romer



# Insurance Motives Explaining Identified Monetary Policy Shocks

| | RR shock (pp) | | | AD shock (pp) | | |
|---|---|---|---|---|---|---|
| | (1) RR Shock | (2) RR Shock | (3) RR Shock | (4) AD Shock | (5) AD Shock | (6) AD Shock |
| BoR Inflation Up | 0.19*** | | 0.094 | 0.055*** | | 0.030* |
| | (3.17) | | (1.58) | (3.10) | | (1.79) |
| BoR Growth Down | -0.19** | | -0.089 | -0.023 | | 0.016 |
| | (-2.35) | | (-1.14) | (-1.15) | | (0.88) |
| BoR Inflation Down | -0.17 | | -0.12 | -0.0060 | | 0.014 |
| | (-1.07) | | (-0.80) | (-0.13) | | (0.32) |
| BoR Growth Up | -0.20 | | -0.31** | 0.013 | | -0.020 |
| | (-1.56) | | (-2.49) | (0.31) | | (-0.56) |
| InsurShr (Cur, Infl) | | 0.56*** | 0.53*** | | 0.17*** | 0.16*** |
| | | (4.76) | (4.58) | | (5.98) | (5.73) |
| InsurShr (Cur, Grow) | | -0.35*** | -0.32*** | | -0.089*** | -0.099*** |
| | | (-4.30) | (-3.47) | | (-3.25) | (-3.34) |
| InsurShr (Cur, Oth) | | 0.13* | 0.11 | | 0.011 | 0.0069 |
| | | (1.97) | (1.63) | | (0.51) | (0.33) |
| $\bar{R}^2$ | 0.099 | 0.21 | 0.23 | 0.058 | 0.20 | 0.20 |
| N | 201 | 200 | 200 | 207 | 206 | 206 |

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# Panel Regression Model

Member-level analysis: $y_{it}$ is a preference measure for member $i$ at meeting $t$.

$$y_{it} = \alpha_i + \delta_t + \sum_{x \in \{U,B,N\}} \gamma_x^I \mathbb{1}\left(BoR_{it}^I = x\right) + \sum_{x \in \{U,B,N\}} \gamma_x^G \mathbb{1}\left(BoR_{it}^G = x\right) + \varepsilon_{it}$$

Meeting fixed effects $\delta_t$ absorb time trends and unobserved aggregate shocks.

Member fixed effects $\alpha_i$ control for fixed preferences (hawk/dove heterogeneity).

Errors clustered at meeting level.

# Preference Variables

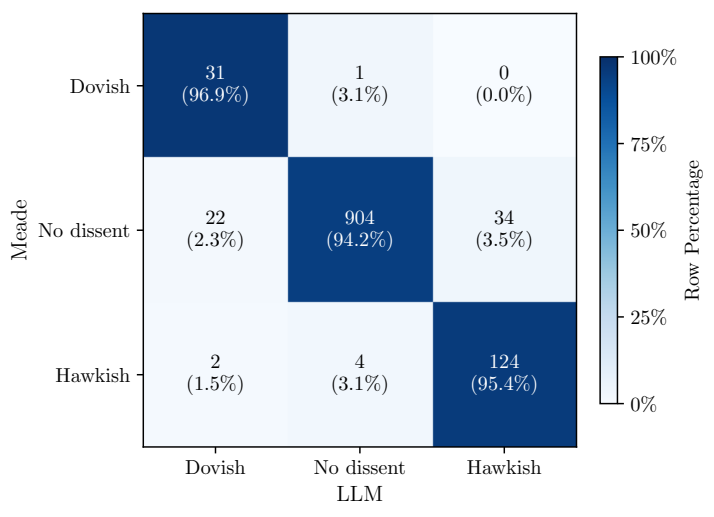
Official voting record of the FOMC has little heterogeneity.

Meade (2005) shows that dissent in voice is more common than dissent in voting and builds a human-coded dataset of individual FFR preferences dataset covering 1989–1997.

We extend the dataset to 1982–2008 using an LLM prompt applied to member statements from FOMC2 that classifies dissent in voice from the FFR decision:

- Dovish dissent
- No dissent
- Hawkish dissent

# Dissent in Voice: LLM vs Meade on Comparable Sample



# Balance of Risk and Individual Member Policy Preferences

| | Meade (ΔFFR bps) | | | LLM Action $(-1, 0, +1)$ | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | 1989–1997 | | | 1982–2019 | | |
| BoR Inflation Up | 5.47*** | 3.25** | 1.69 | 0.28*** | 0.17*** | 0.10*** |
| | (4.33) | (2.29) | (1.15) | (10.32) | (6.57) | (3.96) |
| BoR Growth Down | -3.21** | -0.58 | 0.78 | -0.17*** | -0.08*** | -0.06*** |
| | (-2.40) | (-0.50) | (0.63) | (-6.54) | (-3.54) | (-2.61) |
| BoR Inflation Up × Insurance | | | 8.60*** | | | 0.35*** |
| | | | (3.18) | | | (8.54) |
| BoR Growth Down × Insurance | | | -7.35*** | | | -0.18*** |
| | | | (-3.26) | | | (-4.61) |
| Balanced/ND dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| Insurance dummy | No | No | Yes | No | No | Yes |
| Meeting FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Member FE | No | Yes | Yes | No | Yes | Yes |
| $\bar{R}^2$ | 0.69 | 0.75 | 0.76 | 0.17 | 0.29 | 0.32 |
| N | 1119 | 1119 | 1116 | 4940 | 4935 | 4935 |

# SEP Fed Funds Rate Forecast

Starting in 2012, the SEP added members' preferences on the path of future interest rates.

Match coded views $BoR_{it}^{v}$ to SEP responses at the meeting–speaker level.

Dependent variable $y_{it}$: member $i$'s preferred $FFR_{it}^{h}$ at horizon $h$.

Control for individual growth/inflation forecasts (one-year-ahead GDP and core PCE) and individual uncertainty relative to last 20 years.

Helps isolate perceptions about higher moments from those about first moments.

# Balance of Risk and Individual Member SEP Interest Rate Projections

| | FFR forecast (t+1y) | | | FFR forecast (t+2y) | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| BoR Inflation Up | 0.69*** | 0.34*** | 0.20** | 0.88*** | 0.44*** | 0.30** |
| | (6.67) | (3.73) | (2.20) | (6.44) | (3.35) | (2.34) |
| BoR Growth Down | -0.21 | -0.04 | -0.00 | -0.37* | -0.08 | 0.00 |
| | (-1.67) | (-0.50) | (-0.01) | (-1.85) | (-0.57) | (0.01) |
| Balanced/ND dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| SEP fcast + uncertainty | No | No | Yes | No | No | Yes |
| Meeting FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Member FE | No | Yes | Yes | No | Yes | Yes |
| $\bar{R}^2$ | 0.78 | 0.86 | 0.89 | 0.60 | 0.74 | 0.78 |
| N | 556 | 553 | 552 | 556 | 553 | 552 |

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# Beyond policy actions: Forward-looking policy tilts

So far the focus has been on policy *actions*: FFR target changes and QE actions at the ZLB.

But the stance is not just the current action — it is also the future policy path: policy *tilt*.

We analyse the forward-looking *tilt* in FOMC deliberations.

BoR drives the *tilt*, and the decision whether to signal, especially when there is no current action.

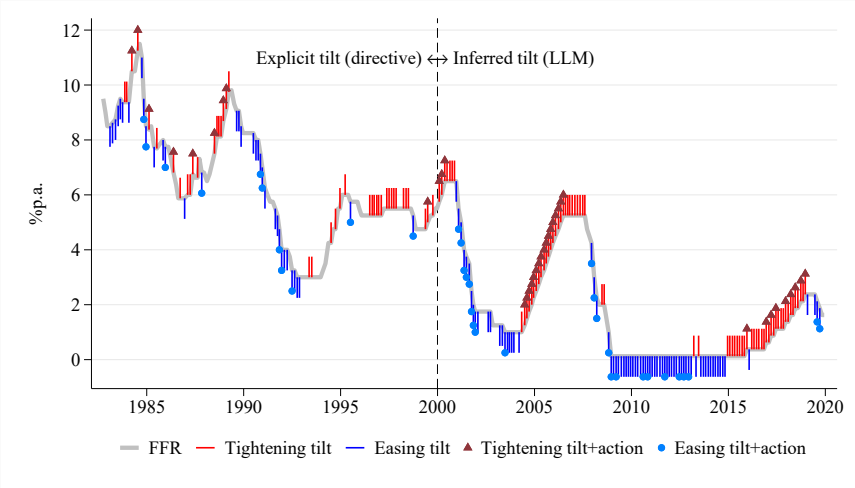
# Measuring forward-looking policy tilt

An explicit policy tilt appears in the FOMC directive from 1983 to 1999 (Thornton–Wheelock, 2000); after 1999 it is replaced by the BoR assessment in the statement, with no explicit tilt.

It was valuable enough to market participants to be a source of leaks before 1994.

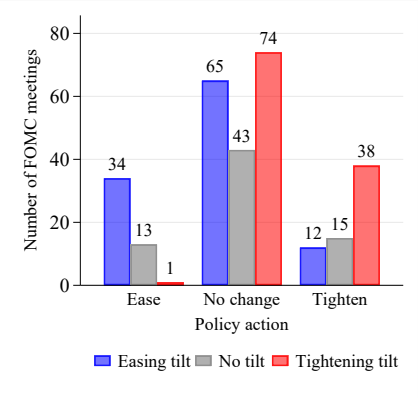
We use an LLM to extend the series beyond 1999, extracting the forward-looking policy preference the FOMC signals about the future path, *beyond the current policy action*.

# Forward-looking FOMC stance as policy instrument: Tilt



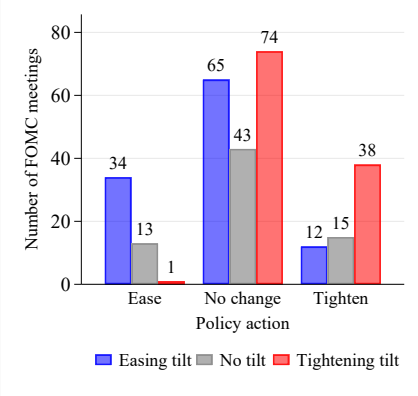
# Wait and signal: Tilt most used when no policy action

A. Policy tilt conditional on current action

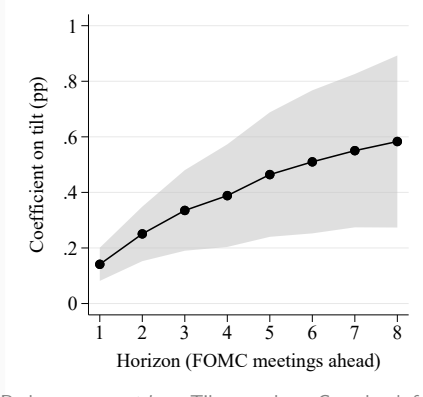


# Wait and signal: Tilt most used when no policy action

A. Policy tilt conditional on current action




B. Tilt predicts future policy path

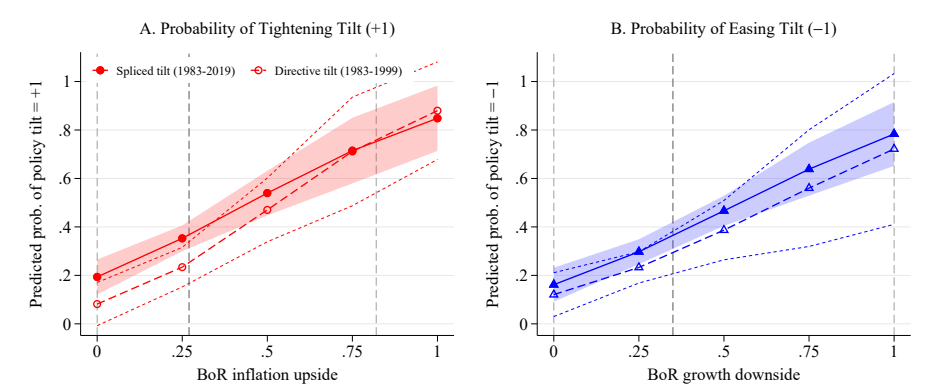



$b_h$: FFR change $t$ to $t+h$ on Tilt$_t$, cond. on Greenbook forecasts and $FFR_t$; 95% HAC band

# Balance of Risk predicts policy tilt

Ordered probit: predicted tilt probabilities

**solid**: spliced tilt 1983–2019 (95% bands); **dashed**: directive tilt 1983–1999




Controls: Greenbook forecasts, lagged policy rates, current action ($\Delta_{1d}FFR$); all four BoR components included

# FOMC actively chooses whether to signal

Since 1994 the FOMC has explicitly communicated its views via the policy statement.

Signals about future policy come at different strengths: the communicated BoR, the policy tilt, and explicit forward guidance in its various forms.

We human-code the signals about future policy in FOMC statements.

# Balance of Risk predicts tilt signal when there is no action (1994–2019)

Dependent var: Change in signal of future policy in FOMC statement $(-1, 0, +1)$

| | All meetings | | | By current action type | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | Ease | **None** | Hike |
| BoR Inflation Up | 1.709*** | 1.513*** | 1.433*** | 0.563 | 1.413*** | 1.120 |
| | (7.77) | (4.00) | (4.21) | (1.07) | (2.98) | (1.35) |
| BoR Growth Down | -1.359*** | -0.886* | -1.191** | -0.219 | -1.221** | -0.613 |
| | (-5.22) | (-1.79) | (-2.60) | (-0.22) | (-2.09) | (-0.78) |
| Action ($\Delta_{1d}FFR$) | | 1.286*** | 1.099** | | | |
| | | (3.32) | (2.49) | | | |
| GB controls | No | No | Yes | Yes | Yes | Yes |
| $\bar{R}^2$ | 0.39 | 0.44 | 0.49 | -0.084 | 0.37 | 0.25 |
| Pseudo $R^2$ | 0.23 | 0.28 | 0.34 | 0.43 | 0.26 | 0.33 |
| N | 158 | 158 | 158 | 28 | 93 | 37 |

The FOMC signals a tightening tilt when it assesses heightened inflation upside risk — a promise to move policy *should the need arise*, without acting yet.

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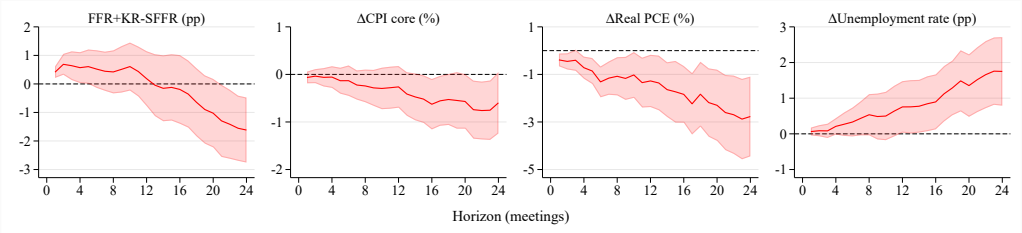
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# BoR Inflation Upside: Insurance premium

$$\text{LP: } \Delta y_{t,t+h} = \alpha_h + \beta_h' BoR_t + \delta_h' BoR_{t-1} + \gamma_h' X_t + \theta_h'(y_t, y_{t-1})' + \varepsilon_{t+h}$$

Effect of BoR Inflation Upside (90% CI)



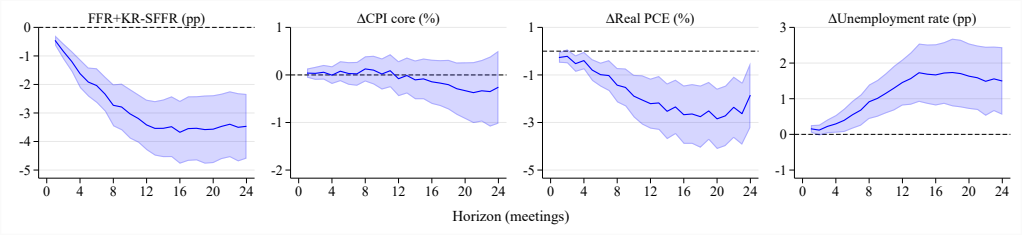

Sample 1982–2019, $h = 1, \ldots, 24$ meetings; controls: Greenbook forecasts, trend inflation, day-before FFR/SSR, yield slope, stock market, lagged BoR and outcome

The FOMC neutralizes the upper tail at some real activity cost — an insurance premium ($1\sigma$: $-0.6\%$ real PCE at 3y).

# BoR Growth Downside: Partial insurance

LP: $\Delta y_{t,t+h} = \alpha_h + \beta_h' BoR_t + \delta_h' BoR_{t-1} + \gamma_h' X_t + \theta_h'(y_t, y_{t-1})' + \varepsilon_{t+h}$

Effect of BoR Growth Downside (90% CI)




Sample 1982–2019, $h = 1, \ldots, 24$ meetings; controls: Greenbook forecasts, trend inflation, day-before FFR/SSR, yield slope, stock market, lagged BoR and outcome

Easing is aggressive and persistent (plateau $-3.5$ pp), yet the downturn is only partially offset: unemployment still rises $\sim$1.5 pp.

"Pushing on a string": the growth tail lies outside the Fed's control and easing is constrained by the ELB.

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# Conclusion

Do policymakers respond to risk/uncertainty?

$\Longrightarrow$ **Yes** (both at committee and individual levels)

$\Longrightarrow$ certainty equivalence AND **insurance**

**Key Takeaway**

The risk management approach has played an important role in the Fed's monetary policy actions and communication.